

2. Safety margin for remaining wealth at the planning horizon, and
3. Allowed probability for falling below the safety margin by the planning horizon.

A more conservative retiree would choose:

- A lower probability for **X%** in order to better avoid the bad outcome,
- A higher threshold for **\$Y** in order to preserve more wealth as a safety margin to cover contingencies, including the possibility of outliving the planning horizon, and
- A longer retirement horizon for **Z** in order to plan for a time period with a smaller chance to outlive.

These three variables are chosen jointly, leading to an overall level of risk for the spending plan. If each parameter is chosen to be extra cautious, the overall degree of plan conservativeness will be high, with the implication that retirement spending must be less in order to meet the requirements of the downside risk threshold. Not all variables need to be overly conservative. For instance, a higher value for **\$Y** means that there is still a larger safety margin for wealth to cover unexpected longevity, which in turn could justify a smaller value for **Z**. Alternatively, if **X%** is already quite low, then perhaps there is more discretion to justify a lower value for **\$Y** or a lower number of years for **Z**.

Ultimately, individuals must find a formulation that creates an acceptable level of downside risk—making each plan comparable—so that the focus can then be shifted to evaluating the overall distribution of spending, the direction of spending over retirement, and the degree of spending volatility. Factors to consider in choosing parameters for the PAY Rule include flexibility to reduce spending, fear of outliving the planning horizon, legacy goals, and the availability of other contingency assets or insurance to protect from spending shocks.

### **Direction of Spending**

After calibrating different spending strategies with the PAY Rule, we can compare the distributions of outcomes using different performance metrics. First, for the direction of spending, different strategies will lead to different